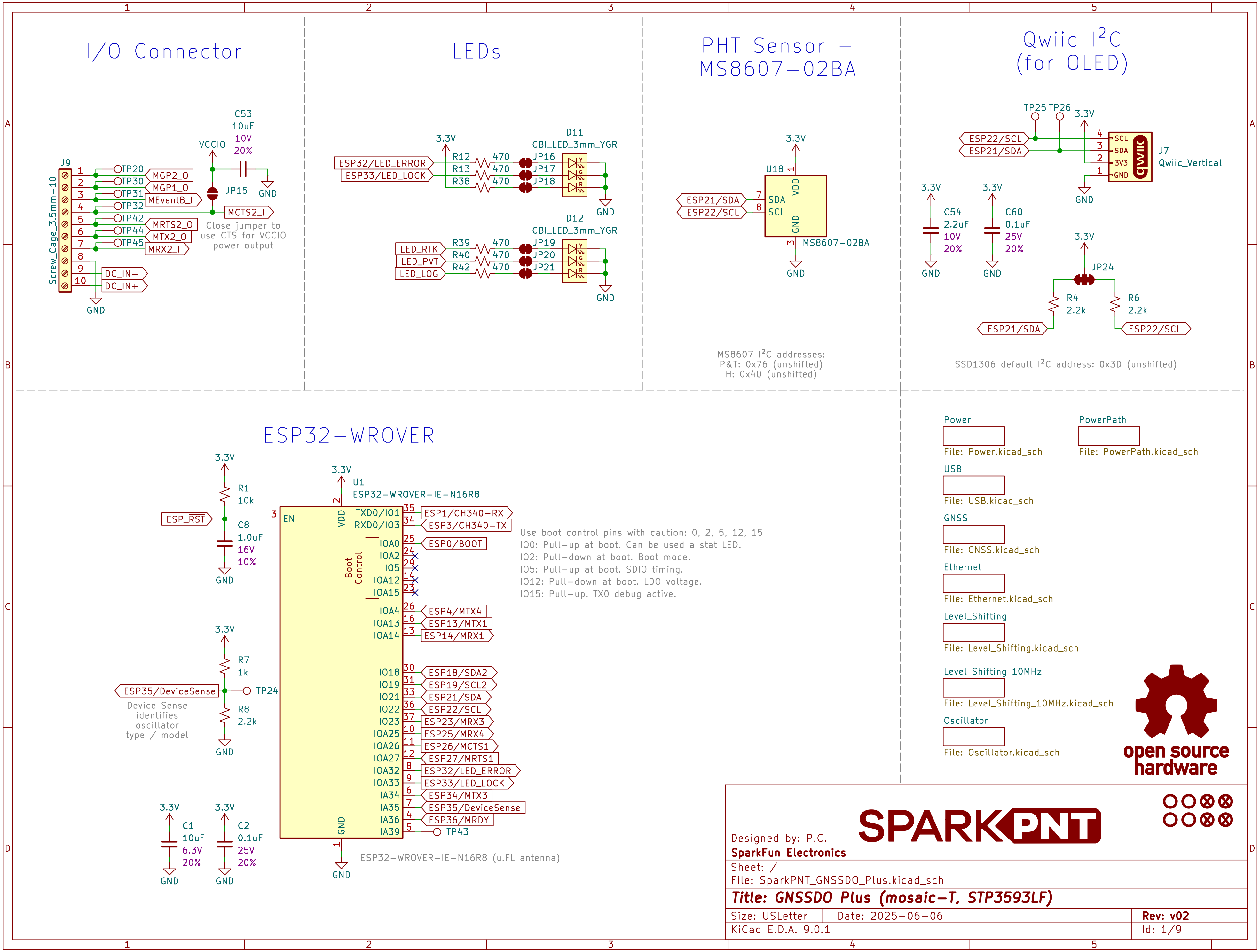
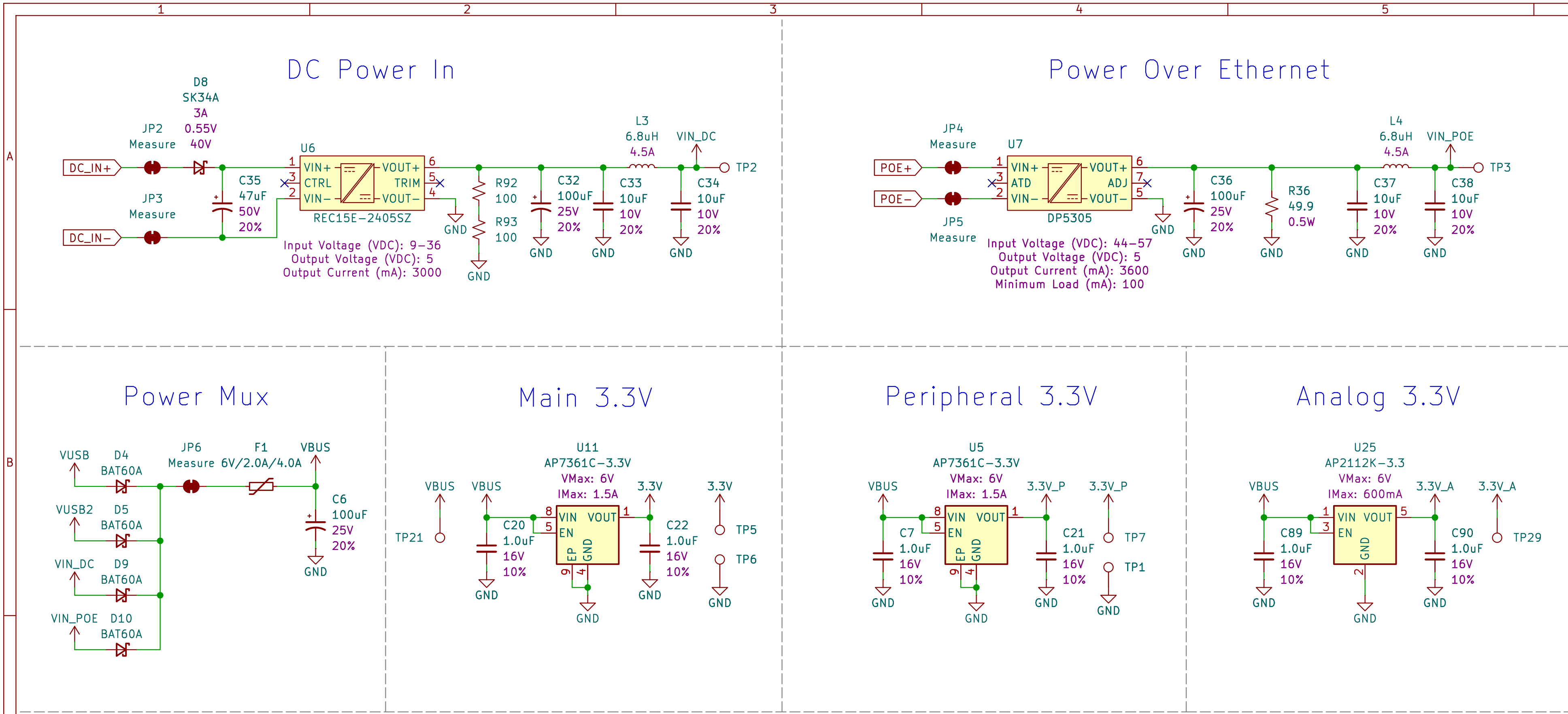


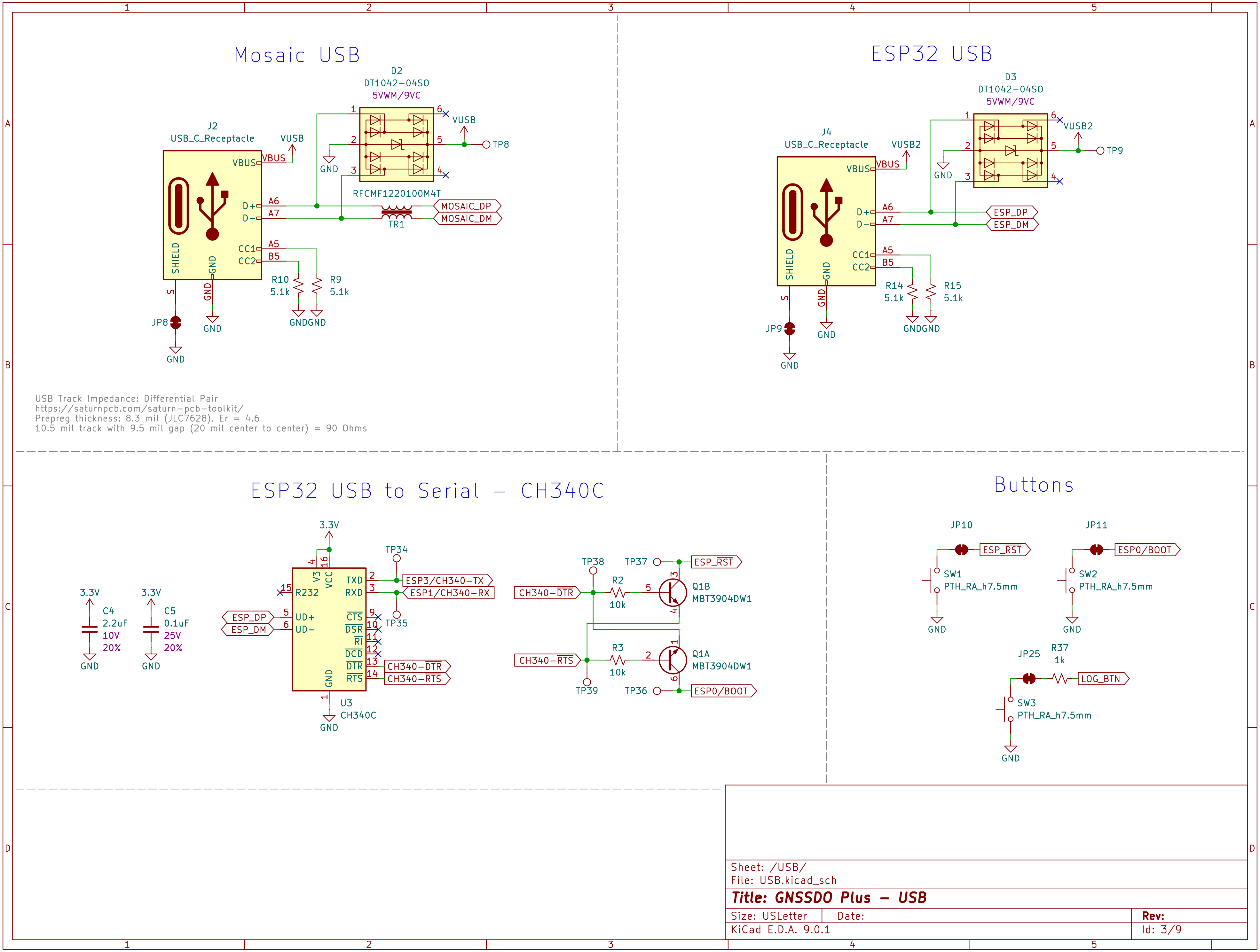
I/O Connector
J9
Screw_Cage_3.5mm-10
TP20
TP30
TP31
TP32
TP42
TP44
TP45
MGP2_O
MGP1_O
MEventB_I
MCTS2_I
MRTS2_O
MTX2_O
MRX2_I
DC_IN-
DC_IN+
GND
VCCIO
C53
10uF
10V
20%
JP15
Close jumper to use CTS for VCCIO power output
LEDs
3.3V
ESP32/LED_ERROR
ESP33/LED_LOCK
R12 470
R13 470
R38 470
JP16
JP17
JP18
D11
CBI_LED_3mm_YGR
LED_RTK
LED_PVT
LED_LOG
R39 470
R40 470
R42 470
JP19
JP20
JP21
D12
CBI_LED_3mm_YGR
PHT Sensor - MS8607-02BA
U18
MS8607-02BA
ESP21/SDA
ESP22/SCL
SDA
SCL
VDD
GND
MS8607 I²C addresses:
P&T: 0x76 (unshifted)
H: 0x40 (unshifted)
Qwiic I²C (for OLED)
TP25 TP26
ESP22/SCL
ESP21/SDA
J7
Qwiic_Vertical
C54
2.2uF
10V
20%
C60
0.1uF
25V
20%
JP24
R4
2.2k
R6
2.2k
SSD1306 default I²C address: 0x3D (unshifted)
ESP32-WROVER
U1
ESP32-WROVER-IE-N16R8
R1
10k
ESP_RST
EN
C8
1.0uF
16V
10%
VDD
TXD0/IO1
RXD0/IO3
IOA0
IOA2
IO5
IOA12
IOA15
Boot Control
IOA4
IOA13
IOA14
IO18
IO19
IO21
IO22
IO23
IOA25
IOA26
IOA27
IOA32
IOA33
IA34
IA35
IA36
IA39
ESP1/CH340-RX
ESP3/CH340-TX
ESP0/BOOT
ESP4/MTX4
ESP13/MTX1
ESP14/MRX1
ESP18/SDA2
ESP19/SCL2
ESP21/SDA
ESP22/SCL
ESP23/MRX3
ESP25/MRX4
ESP26/MCTS1
ESP27/MRTS1
ESP32/LED_ERROR
ESP33/LED_LOCK
ESP34/MTX3
ESP35/DeviceSense
ESP36/MRDY
TP43
Use boot control pins with caution: 0, 2, 5, 12, 15
IO0: Pull-up at boot. Can be used a stat LED.
IO2: Pull-down at boot. Boot mode.
IO5: Pull-up at boot. SDIO timing.
IO12: Pull-down at boot. LDO voltage.
IO15: Pull-up. TX0 debug active.
R7
1k
ESP35/DeviceSense
TP24
R8
2.2k
Device Sense identifies oscillator type / model
C1
10uF
6.3V
20%
C2
0.1uF
25V
20%
ESP32-WROVER-IE-N16R8 (u.FL antenna)
Power
File: Power.kicad_sch
PowerPath
File: PowerPath.kicad_sch
USB
File: USB.kicad_sch
GNSS
File: GNSS.kicad_sch
Ethernet
File: Ethernet.kicad_sch
Level_Shifting
File: Level_Shifting.kicad_sch
Level_Shifting_10MHz
File: Level_Shifting_10MHz.kicad_sch
Oscillator
File: Oscillator.kicad_sch
open source hardware
SPARKPNT
Designed by: P.C.
SparkFun Electronics
Sheet: /
File: SparkPNT_GNSSDO_Plus.kicad_sch
Title: GNSSDO Plus (mosaic-T, STP3593LF)
Size: USLetter
Date: 2025-06-06
Rev: v02
KiCad E.D.A. 9.0.1
Id: 1/9



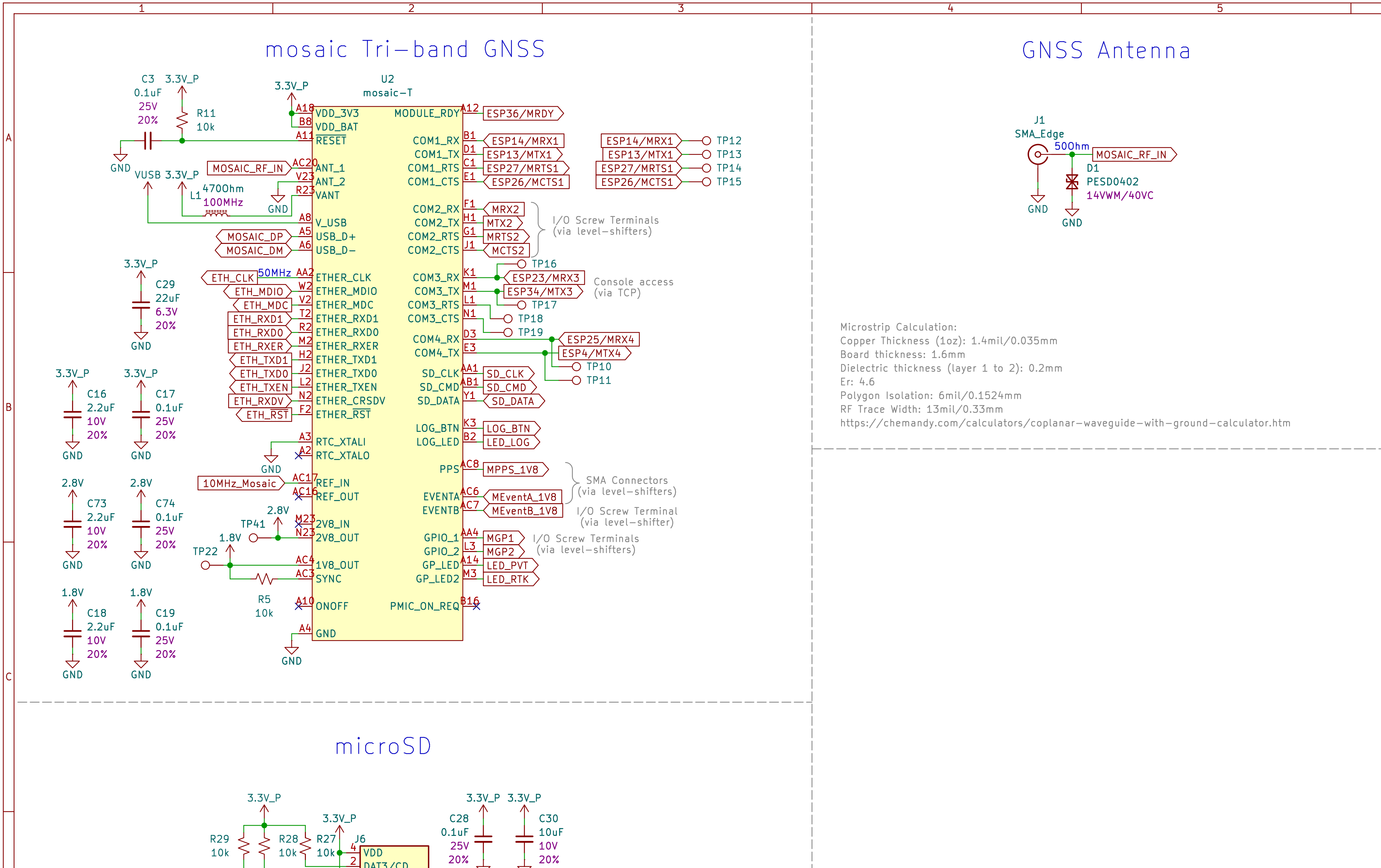
Sheet: /Power/
File: Power.kicad_sch

Title: GNSSDO Plus − Power

Size: USLetter | Date: | **Rev:**
KiCad E.D.A. 9.0.1 | Id: 2/9



Mosaic USB
D2
DT1042-04SO
5VWM/9VC
J2
USB_C_Receptacle
VUSB
VBUS
D+
D-
A6
A7
CC1
CC2
A5
B5
SHIELD
GND
S
JP8
R10
5.1k
R9
5.1k
GNDGND
TP8
RFCMF1220100M4T
TR1
MOSAIC_DP
MOSAIC_DM
USB Track Impedance: Differential Pair
https://saturnpcb.com/saturn-pcb-toolkit/
Prepreg thickness: 8.3 mil (JLC7628). Er = 4.6
10.5 mil track with 9.5 mil gap (20 mil center to center) = 90 Ohms
ESP32 USB
D3
DT1042-04SO
5VWM/9VC
J4
USB_C_Receptacle
VUSB2
VBUS
TP9
ESP_DP
ESP_DM
R14
5.1k
R15
5.1k
JP9
GND
ESP32 USB to Serial - CH340C
3.3V
C4
2.2uF
10V
20%
C5
0.1uF
25V
20%
TP34
TP35
R232
V3
VCC
TXD
RXD
UD+
UD-
CTS
DSR
RI
DCD
DTR
RTS
U3
CH340C
ESP3/CH340-TX
ESP1/CH340-RX
CH340-DTR
CH340-RTS
TP38
TP37
TP39
TP36
R2
10k
R3
10k
Q1B
MBT3904DW1
Q1A
MBT3904DW1
ESP_RST
ESP0/BOOT
Buttons
JP10
JP11
SW1
PTH_RA_h7.5mm
SW2
PTH_RA_h7.5mm
JP25
R37
1k
LOG_BTN
SW3
PTH_RA_h7.5mm
Sheet: /USB/
File: USB.kicad_sch
Title: GNSSDO Plus - USB
Size: USLetter
Date:
Rev:
KiCad E.D.A. 9.0.1
Id: 3/9

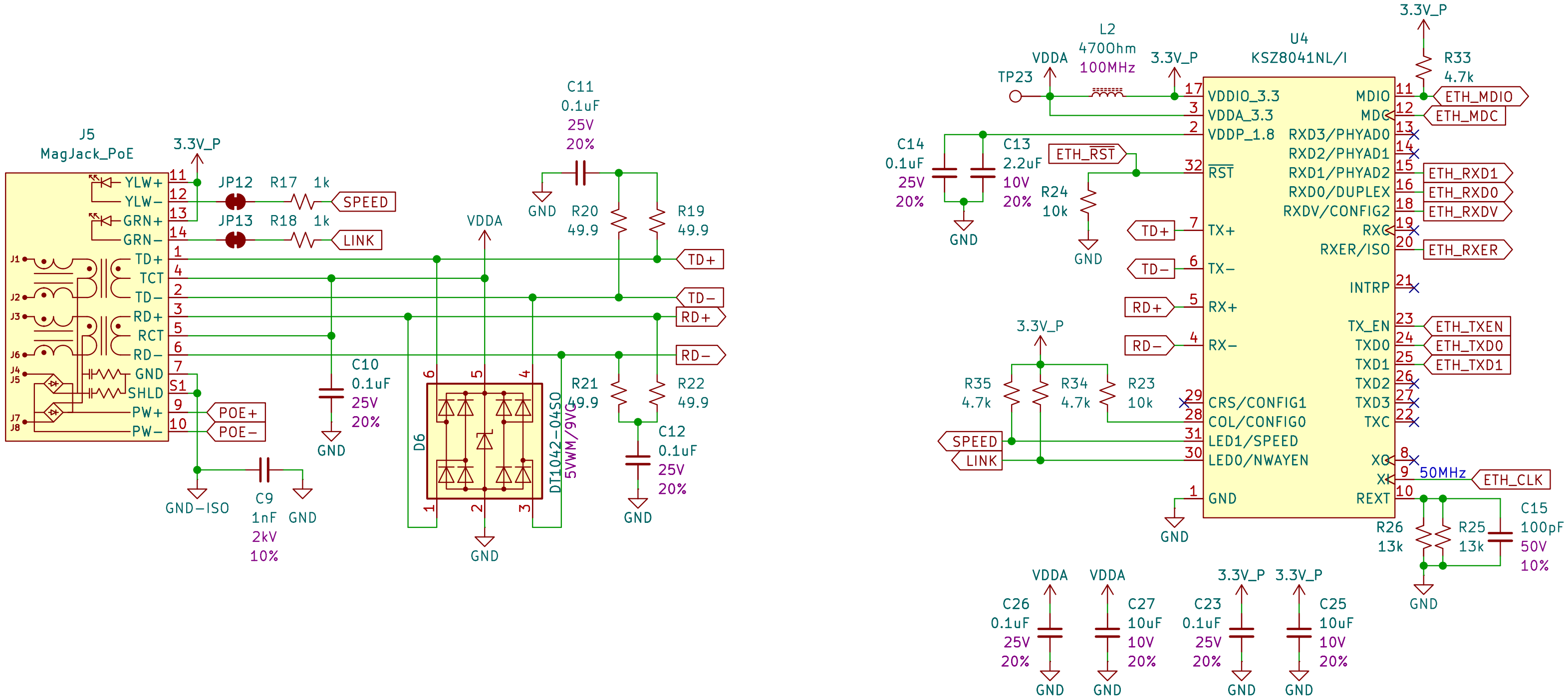


Sheet: /GNSS/
File: GNSS.kicad_sch

Title: GNSSDO Plus - GNSS

Size: USLetter Date: Rev:
KiCad E.D.A. 9.0.1 Id: 4/9

Ethernet



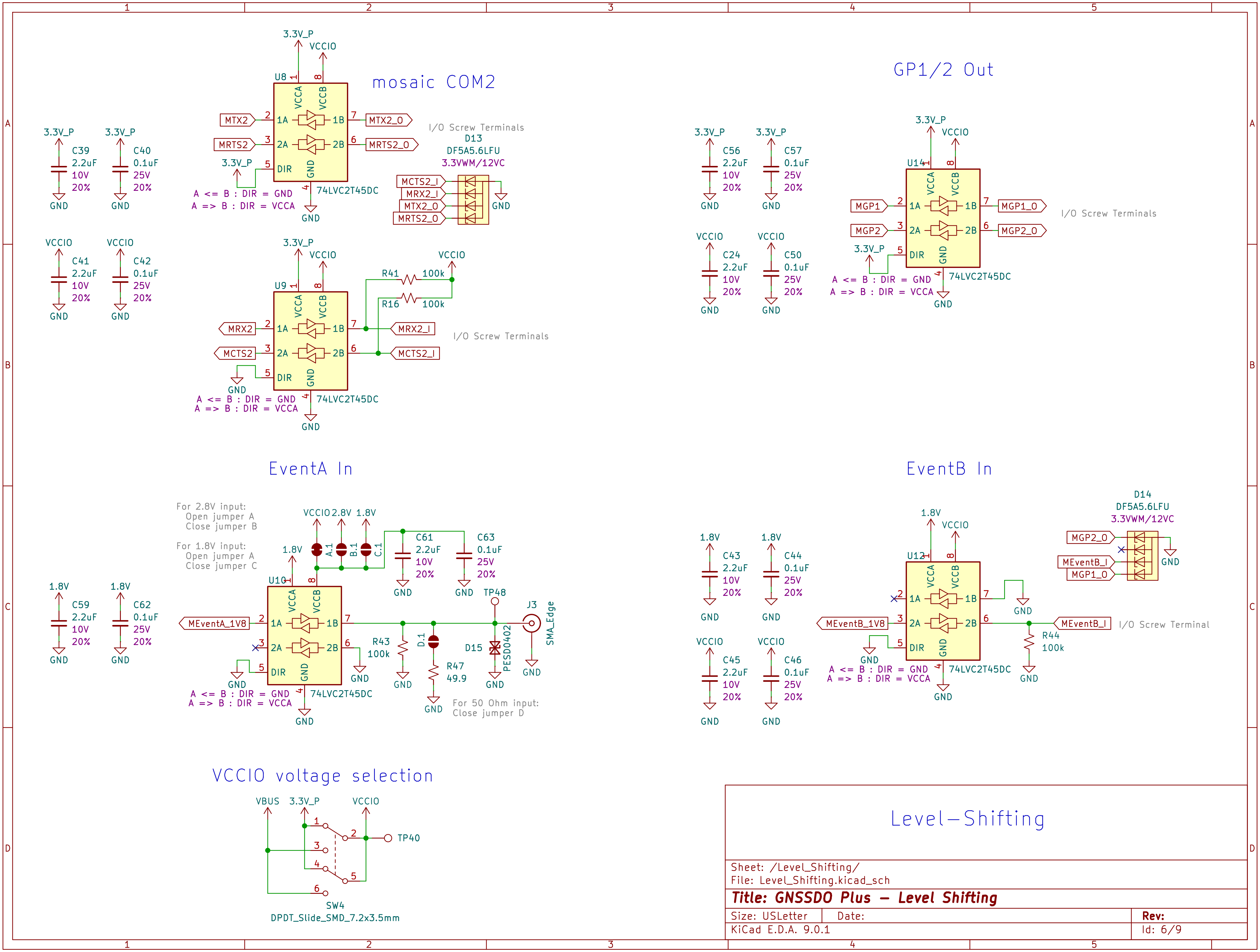
Ethernet Track Impedance: Differential Pair
https://saturnpcb.com/saturn-pcb-toolkit/
Prepreg thickness: 8.3 mil (JLC7628). Er = 4.6
9.0 mil track with 11.0 mil gap (20 mil center to center) = 100 Ohms
Each pair should match in length to better than 0.5mm

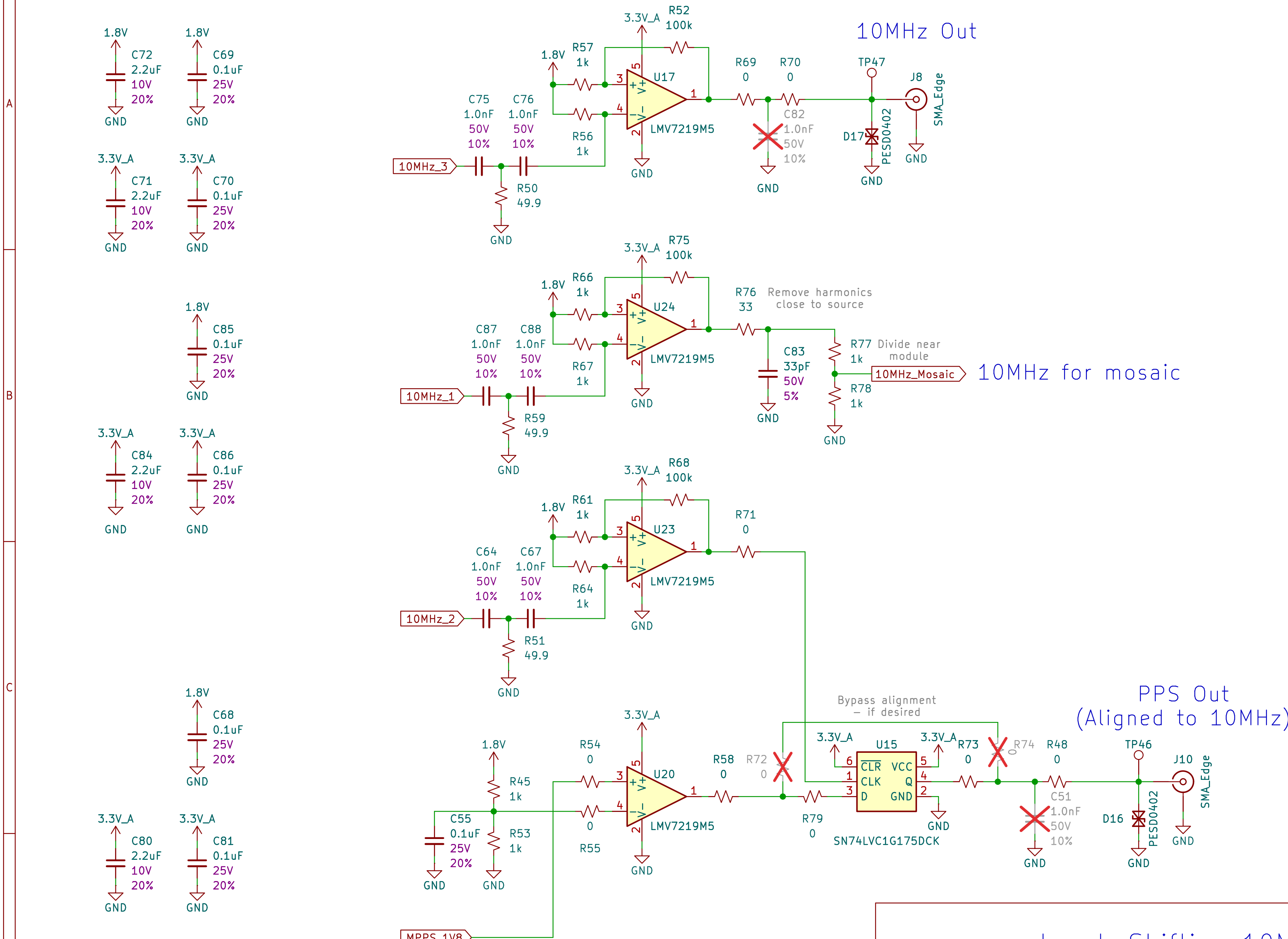
Sheet: /Ethernet/
File: Ethernet.kicad_sch

Title: GNSSDO Plus – Ethernet

Size: USLetter | Date: | Rev:

KiCad E.D.A. 9.0.1 | Id: 5/9


mosaic COM2
U8 74LVC2T45DC
3.3V_P VCCIO
MTX2 MTX2_O
MRTS2 MRTS2_O
A <= B : DIR = GND
A => B : DIR = VCCA
I/O Screw Terminals
D13 DF5A5.6LFU 3.3VWM/12VC
MCTS2_I MRX2_I MTX2_O MRTS2_O
U9 74LVC2T45DC
R41 100k R16 100k
MRX2 MRX2_I
MCTS2 MCTS2_I
C39 2.2uF 10V 20%
C40 0.1uF 25V 20%
C41 2.2uF 10V 20%
C42 0.1uF 25V 20%
GP1/2 Out
U14 74LVC2T45DC
MGP1 MGP1_O
MGP2 MGP2_O
C56 2.2uF 10V 20%
C57 0.1uF 25V 20%
C24 2.2uF 10V 20%
C50 0.1uF 25V 20%
EventA In
For 2.8V input: Open jumper A Close jumper B
For 1.8V input: Open jumper A Close jumper C
VCCIO 2.8V 1.8V
A.1 B.1 C.1 D.1
C61 2.2uF 10V 20%
C63 0.1uF 25V 20%
U10 74LVC2T45DC
MEventA_1V8
R43 100k
R47 49.9
D15 PESD0402
TP48 J3 SMA_Edge
For 50 Ohm input: Close jumper D
C59 2.2uF 10V 20%
C62 0.1uF 25V 20%
EventB In
D14 DF5A5.6LFU 3.3VWM/12VC
MGP2_O MEventB_I MGP1_O
U12 74LVC2T45DC
MEventB_1V8 MEventB_I
R44 100k
I/O Screw Terminal
C43 2.2uF 10V 20%
C44 0.1uF 25V 20%
C45 2.2uF 10V 20%
C46 0.1uF 25V 20%
VCCIO voltage selection
VBUS 3.3V_P VCCIO
TP40
SW4 DPDT_Slide_SMD_7.2x3.5mm
Level-Shifting
Sheet: /Level_Shifting/
File: Level_Shifting.kicad_sch
Title: GNSSDO Plus - Level Shifting
Size: USLetter Date:
Rev:
KiCad E.D.A. 9.0.1
Id: 6/9



Note: all paths use identical components to
ensure equal temperature dependence

Level-Shifting 10MHz

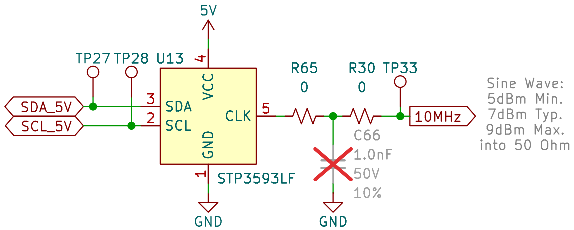
Sheet: /Level_Shifting_10MHz/
File: Level_Shifting_10MHz.kicad_sch

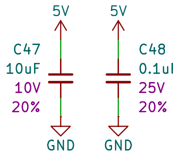
Title: GNSSDO Plus - Level Shifting 10MHz

Size: USLetter	Date:	Rev:
KiCad E.D.A. 9.0.1		Id: 7/9

10MHz Oscillator – STP3593LF

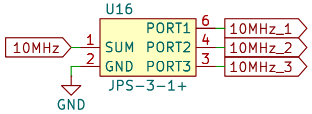
Supply Voltage: 5.0V (4.75V Min., 5.25V Max.)
Current Consumption: 1500mA (Warm Up), 600mA (Steady State)



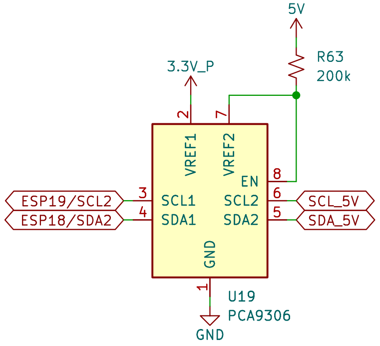


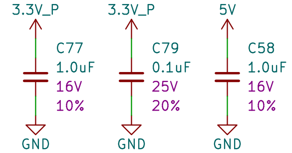
3-Way Splitter – JPS-3-1+

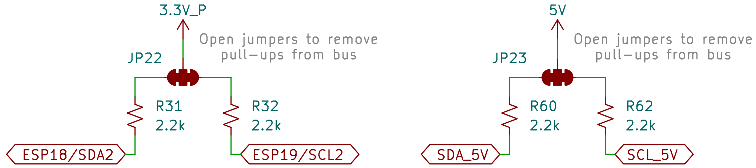
Typical Total Loss: 5.0dB at 10MHz



I^{2}C Level Shifting – PCA9306



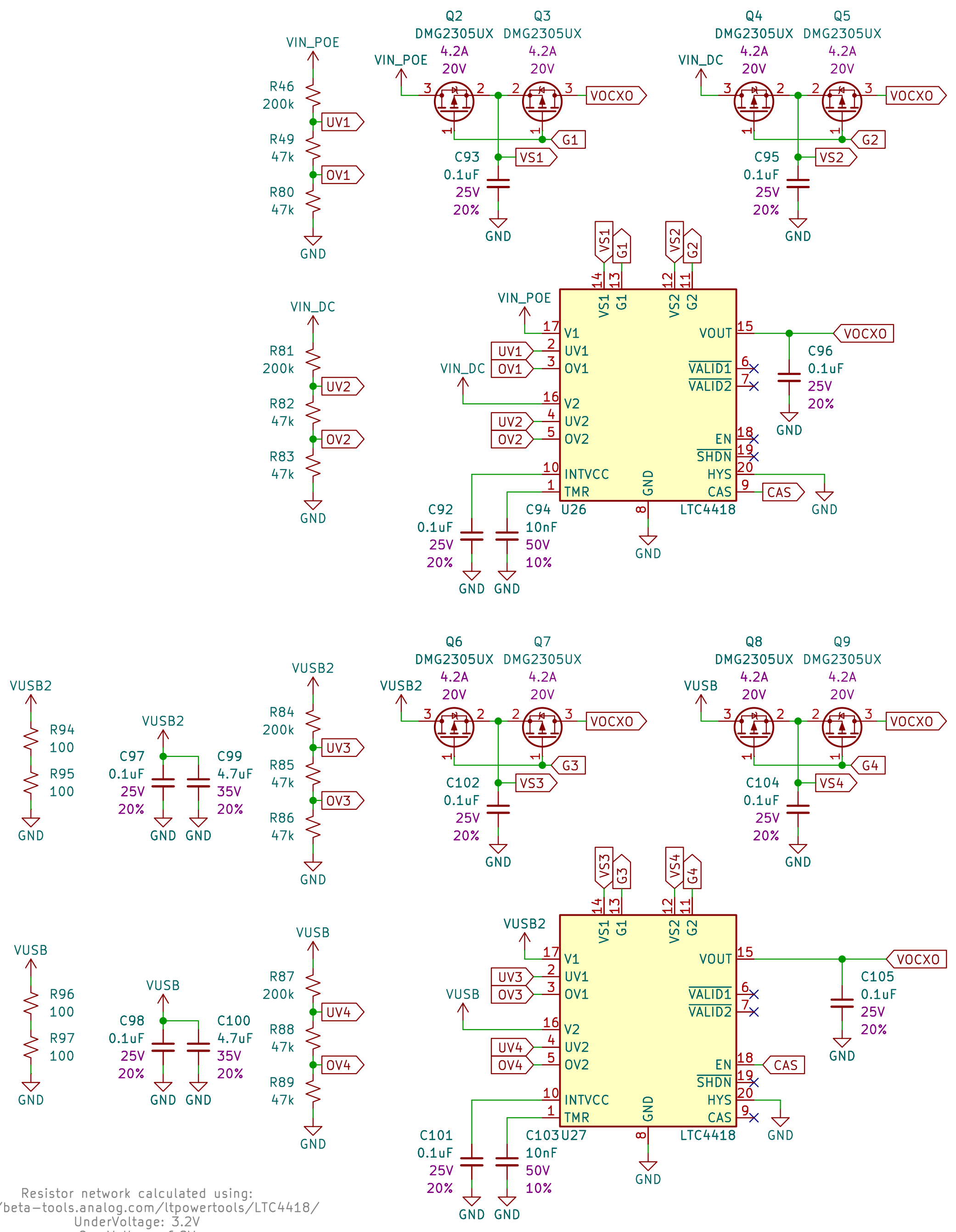


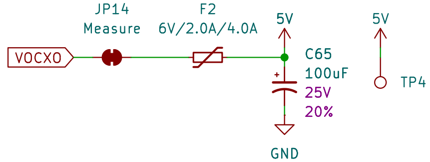


Sheet: /Oscillator/
File: Oscillator.kicad_sch

Title: GNSSDO Plus – Oscillator

Size: A4	Date:	**Rev:**
KiCad E.D.A. 9.0.1		Id: 8/9





PowerPath Priority:
PoE
VIN
VUSB2 (CONFIG ESP32)
VUSB (CONFIG MOSAIC)

VUSB2 is prioritized over VUSB as: CONFIG MOSAIC is more likely to be used for USB data and connected to a USB 2.0 port; CONFIG ESP32 is less likely to be used for USB data and easier to connect to a Power Delivery supply.

Resistor network calculated using:
http://beta-tools.analog.com/ltpowertools/LTC4418/
UnderVoltage: 3.2V
OverVoltage: 6.2V

OCXO Power Mux –
PowerPath Controllers LTC4418

Sheet: /PowerPath/
File: PowerPath.kicad_sch

Title: GNSSDO Plus – OCXO Power Mux (PowerPath)

Size: A4	Date:	Rev:
KiCad E.D.A. 9.0.1		Id: 9/9